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Technical paper

A digital twin calibration for an automated material handling system in a semiconductor fab

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ABSTRACT

To address the complex, dynamic, and stochastic nature of an automated material handling system (AMHS) in a semiconductor fabrication facility (fab), practitioners have used a high-fidelity discrete-event simulation as its digital twin model for decision-making over several decades. Previous studies have focused on fast digital twin-based decision-making in AMHSs under the assumption that their digital twin models are credible enough to prescribe decisions. However, parameter uncertainty and intrinsic bias in an AMHS digital twin model can lead to an inaccurate representation of its field system. To address the challenge, this paper introduces the Bayesian calibration, which modularly estimates a digital twin outcome and its discrepancy using Gaussian process priors. A calibration framework for digital twin-based decision-making is also presented using an AMHS example. Our experimental results with various AMHS operating scenarios demonstrate that: (1) a sophisticated digital twin calibration is necessary, especially when AMHSs operate under heavy-workload scenarios; and (2) exploring model bias considerably decreases the prediction error of an AMHS digital twin within a limited number of field observations. Moreover, we discuss the applicability of the approach to digital twins in various fields.

1. Introduction

The definitions of digital twins differ and have evolved for diverse industries [1,2] while the digital twins for manufacturing and material handling systems can be defined as a one-to-one virtual replica of a field system with real-time bidirectional interaction [3–5]. According to the perception of system dynamics, the digital twins aim to support decision-making by providing automatic process evaluation and optimization over time [1,6]. For sustainable digital twin construction, one of the crucial tasks is to keep a digital twin model consistent with its field system because an inaccurate representation of a digital twin model can lead to unexpected behavior of a field system [7,8].

Automated material handling systems (AMHSs) have been successfully implemented in semiconductor and display fabrication facilities (fabs), significantly contributing to productivity improvements. Since the material handling process in AMHSs involves complex manufacturing steps and control logic, high-fidelity discrete-event simulations (DESs) have served as digital twin models to predict material handling performance under various system configurations [9–15]. Fig. 1 presents a field AMHS system and its digital twin model for an AMHS in a semiconductor fab.

However, similar to other computer models, a digital twin model of an AMHS is subject to issues such as parameter uncertainty and model bias, potentially leading to suboptimal decision-making [16–18]. For example, in an AMHS, delivery time (DT) is a critical metric. If vehicle congestion occurs in the field system, which the digital twin model fails to accurately capture, it can lead to unproductive material handling operations and fail to meet the production targets. To provide an accurate representation of an AMHS digital twin model, performance-level calibration has been applied, focusing on the gap of average values of material handling events between a field system and its digital twin model [7]. Since performance-level calibration aggregates the immediate impact of anomalous events, it provides a steady-state diagnosis of material handling operations for an upcoming production plan to field practitioners.

One common calibration approach in real-world AMHSs is manual calibration, where practitioners iteratively adjust calibration parameters until digital twin outcomes match field observations for an upcoming production plan. However, this method is often computationally expensive for high-fidelity digital twin models. Several domain-specific digital twin calibration approaches have been proposed for traffic simulation in urban transportation; however, these studies primarily

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focus on optimizing calibration parameters without considering model bias [19–25].

A digital twin model still has a large bias from the actual AMHS because it is infeasible to build a practically computable digital twin model incorporating all the AMHS complexities. Therefore, understanding and exploiting bias is crucial for successful digital twin-based decision-making. This paper proposes to simultaneously optimize calibration parameters and compensate for model bias. For rigorous digital twin calibration, we leverage the theory of Bayesian calibration [13]. Our simulation experiments demonstrate that the approach accounts for both parameter uncertainty and systematic discrepancies in an AMHS digital twin. Finally, we discuss its applicability to digital twin models across various domains.

The remainder of this paper is organized as follows. Section 2 reviews the relevant literature and challenges. Section 3 introduces the Bayesian calibration for digital twin models. Section 4 presents a calibration framework for digital twin-based decision-making in an AMHS. Section 5 validates the proposed approach and discusses its applicability. Section 6 concludes and suggests future research directions.

2. Literature review

In a predictive manner, engineers have developed DESs to evaluate facility designs or validate the suitability of novel algorithms in AMHSs because a simulation is much less expensive than setting up a field system and allows testing their effectiveness in specific production environments. Lee et al. [26] suggested a practical routing algorithm to disperse vehicle traffic based on a monitoring system to identify and resolve congestion bottlenecks in an AMHS of a fab. They used a high-fidelity and large-scale simulation from a Korean semiconductor manufacturing company to validate the superiority of the proposed approach. Benzoni et al. [27] proposed heuristic-driven look-ahead dispatching algorithms in an AMHS. They confirmed the effectiveness of the proposed algorithms through simulation experiments based on a real semiconductor fab. Lee et al. [28] presented a learning-based routing algorithm to predict network traffic flow and adjust the routing parameters for critical bottleneck rails in an AMHS. They constructed an AMHS simulation identical to real fab configurations to validate the effectiveness of the presented approach.

In a prescriptive manner, previous studies relied on digital twin-based decision-making in AMHSs. They reported that surrogate models can be suitable when an AMHS simulation requires high computation time. Kang et al. [29] investigated a Gaussian process (GP) based simulation optimization to obtain an efficient vehicle repositioning policy within limited simulation runs. The experimental results suggest that the surrogate model can adaptively estimate the simulation outcomes and derive an efficient solution. Kuo et al. [30] studied a neural network-based surrogate model to accelerate decision-making in an AMHS of a fab. The model can predict the simulation outcomes with limited training data. Can et al. [31] presented a comparative study of two surrogate models to predict the inventory cost in an AMHS: genetic programming and neural networks. Although the genetic programming-based model requires a heavier computational effort, it can provide a high validity with explicit functions for strategic management in the AMHS.

Most studies of digital twin-based optimization in manufacturing and material handling systems assume that a digital twin model can precisely represent an actual response across different control parameters. However, a computer model cannot perfectly represent a complex field system [32]. To evaluate the validity of digital twin models, Mertens et al. [8] proposed a continuous validation workflow including model validation and calibration processes; however, the investigation of a specific calibration methodology was left for future research. Muñoz et al. [7], who introduced a model validity measure based on the periodically captured snapshots of a field system and its digital twin model, proposed an offline validation method that tracks the behaviors of the field system and the model. Lugaresi et al. [33] defined an online digital twin validation at the event and performance levels for production planning and control while a field system is in operation. The proposed validation approach evaluates the validity of the digital twin model by detecting similarities and deviations over time. Although several digital twin validation approaches have been proposed, to the best of our knowledge, a systematic calibration approach for digital twins in manufacturing and material handling systems remains an open challenge.

Kennedy et al. [18] identified the sources of uncertainty in a computer model as parameter uncertainty, model inadequacy, and residual variability. They presented a Bayesian calibration to estimate calibration parameters and compensate for model inadequacy. Parameter uncertainty refers to unknowable parameters in a field system but controllable in digital twin models; quantifying the uncertainty of calibration parameters is critical to reducing the distinction between a field system and its digital twin. Model inadequacy is the difference between the true mean observations of a field system and its computer model with optimal calibration parameters. Residual variability represents stochastic noise. The sometimes called 'Kennedy-and-O'Hagan (KOH) approach' has been tailored to various fields. Higdon et al. [34], who proposed a statistical framework based on the KOH approach, carried out a calibration to account for model inadequacy and adjust calibration parameters in a thermal simulator. According to the experimental results, the proposed framework incorporates model inadequacy in limited simulation runs. Bayarri et al. [35], who provided a model calibration framework based on the KOH approach, applied the approach to an example of a resistance spot welding problem. Chong et al. [36] varied the KOH approach, improving the computational speed with the large dataset for a building energy simulation. The presented approach used an extension of the Hamiltonian Monte Carlo to derive the posterior distributions of uncertain parameters. The experiment results showed that the approach reduced the computation cost while maintaining acceptable prediction accuracy. Wong et al. [37] reported a frequentist framework for model calibration with theoretical proofs based on the KOH approach. They used the bootstrapping method to prevail the uncertainty in the digital model for a fluid dynamics problem.

3. Bayesian calibration for digital twin models

This section introduces the statistical background for the Bayesian calibration of digital twin models and the role of parameter uncertainty and discrepancy. We combine field observations and digital twin outcomes to improve prediction accuracy; this is suitable when field data acquisition is extremely limited but its digital twin model is relatively cheap to evaluate [34].

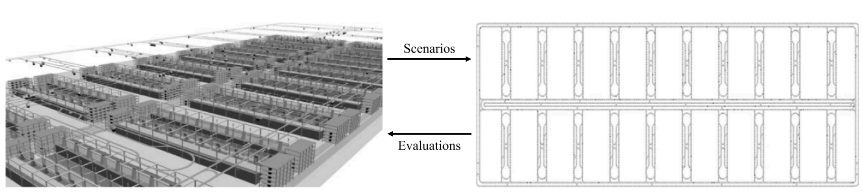

Fig. 1. Example of a field system and its digital twin model for an AMHS in a fab.

3.1. Statistical formulation for Bayesian calibration

Let $\boldsymbol{x} = (x_1, \dots, x_p)$ represent the p control parameters in a field system and its digital twin model. Let $\boldsymbol{t} = (t_1, \dots, t_l)$ represent the l calibration parameters that only exist in digital twin models. Let y denote the key performance metric to be predicted. The relationship between a field observation and its digital twin outcome can be formulated as follows [18]:

$$y(\boldsymbol{x}) = \eta(\boldsymbol{x}, \boldsymbol{t}^*) + \delta(\boldsymbol{x}) + \epsilon(\boldsymbol{x}), \tag{1}$$

where $y(\boldsymbol{x})$ denotes the field observation given $\boldsymbol{x}$, $\eta(\boldsymbol{x}, \boldsymbol{t})$ denotes the digital twin outcome given $\boldsymbol{x}$ and $\boldsymbol{t}$, $\boldsymbol{t}^*$ represents the optimally calibrated values of $\boldsymbol{t}$, $\delta(\boldsymbol{x})$ denotes the systematic discrepancy varying with $\boldsymbol{x}$ between the field observation and the digital twin outcome, and $\epsilon(\boldsymbol{x})$ accounts for an observation error. We assume $\epsilon(\boldsymbol{x})$ to be a zero-mean Gaussian noise with a small standard deviation σ_ϵ for numerical stability [38, 39].

Typically, running a digital twin model to get the outcome $\eta(\boldsymbol{x}, \boldsymbol{t})$ is computationally expensive, so it cannot be evaluated for every possible $\boldsymbol{x}$ and $\boldsymbol{t}$ combination. In practice, the $\eta(\boldsymbol{x}, \boldsymbol{t})$ is replaced with a statistical surrogate estimated with a finite number of digital twin data. A GP surrogate model is popularly used in the Bayesian calibration, because of its capabilities in non-linear/non-parametric modeling and uncertainty quantification [16, 40]. An unknown function such as $\eta(\boldsymbol{x}, \boldsymbol{t})$ is considered as a random realization of a GP, a stochastic process for a finite number of input configurations is jointly distributed according to a multivariate Gaussian distribution [41, 42]. The digital twin model $\eta(\boldsymbol{x}, \boldsymbol{t})$ is specified by a zero mean and a covariance function $k_\eta((\boldsymbol{x}, \boldsymbol{t}), (\boldsymbol{x}', \boldsymbol{t}'))$, written as

$$\eta(\boldsymbol{x}, \boldsymbol{t}) \sim GP(0, k_\eta((\boldsymbol{x}, \boldsymbol{t}), (\boldsymbol{x}', \boldsymbol{t}'))), \tag{2}$$

The role of the covariance function is to quantify the covariance between a pair of different control and calibration parameters used for the digital twin surrogate. In this paper, an anisotropic covariance function is considered a product correlation because the control parameters and the calibration parameters may exhibit different degrees of covariance with the digital twin outcomes. This can extend analytic tractability for managerial findings and provide sophisticated predictions but requires high-dimensional kernel parameter estimation compared to an isotropic kernel

$$k_\eta((\boldsymbol{x}, \boldsymbol{t}), (\boldsymbol{x}', \boldsymbol{t}')) = \alpha_\eta \exp\left\{ -\sum_{k=1}^{p} \beta_{\eta,k} (x_k - x_k')^2 - \sum_{q=1}^{l} \beta_{\eta,p+q} (t_q - t_q')^2 \right\} + \epsilon I, \tag{3}$$

where x_k and x_k' are the k-th element of $\boldsymbol{x}$ and $\boldsymbol{x}'$ respectively, t_q is the q-th element of $\boldsymbol{t}$, the amplitude α_η controls the overall variance, and the dependence strength set $\boldsymbol{\beta}_\eta = (\beta_{\eta,1}, \dots, \beta_{\eta,p}, \beta_{\eta,p+1}, \dots, \beta_{\eta,p+l})$ reflects the comprehensive impact of control and calibration parameters on the digital twin surrogate.

Here, we provide a discrepancy function $\delta(\boldsymbol{x})$ of the digital twin model using a GP prior. $\delta(\boldsymbol{x})$ is specified by a zero mean function and a discrepancy covariance function $k_\delta(\boldsymbol{x}, \boldsymbol{x}')$ as follows:

$$\delta(\boldsymbol{x}) \sim GP(0, k_\delta(\boldsymbol{x}, \boldsymbol{x}')). \tag{4}$$

We consider an anisotropic covariance function because each control parameter may have its own impact on the discrepancy surrogate, as follows:

$$k_\delta(\boldsymbol{x}, \boldsymbol{x}') = \alpha_\delta \exp\left\{ -\sum_{k=1}^{p} \beta_{\delta,k} (x_k - x_k')^2 \right\}, \tag{5}$$

where the amplitude α_δ controls the overall variance and the dependence strength $\boldsymbol{\beta}_\delta = (\beta_{\delta,1}, \dots, \beta_{\delta,p})$ reflects the impact of control parameters on the discrepancy surrogate.

3.2. Statistical estimation and calibration

Let m denote the number of digital twin evaluations. Let $(\boldsymbol{x}_i, \boldsymbol{t}_i)$ denote the control and calibration parameters used for the i-th digital twin evaluation for $i = 1, \dots, m$. Let $\boldsymbol{\eta} = (\eta(\boldsymbol{x}_1, \boldsymbol{t}_1), \dots, \eta(\boldsymbol{x}_m, \boldsymbol{t}_m))^T$ denote the digital twin outcomes. According to the GP surrogate specification (2), we can have the probability of the outcomes given the kernel parameters $(\alpha_\eta, \boldsymbol{\beta}_\eta)$, which gives the likelihood

$$L(\boldsymbol{\eta} | \alpha_\eta, \boldsymbol{\beta}_\eta) \propto |\Sigma_\eta|^{-\frac{1}{2}} \exp\left\{ -\frac{1}{2} \boldsymbol{\eta}^T \Sigma_\eta^{-1} \boldsymbol{\eta} \right\}, \tag{6}$$

where Σ_η is a $m \times m$ covariance matrix with its (i,j)-th element equal to $k_\eta((\boldsymbol{x}_i, \boldsymbol{t}_i), (\boldsymbol{x}_j, \boldsymbol{t}_j))$ and $|\Sigma_\eta|$ is the determinant of Σ_η.

Let n denote the number of field evaluations. Let $\boldsymbol{x}_i^*$ denote the vector of control parameters for the i-th field observation for $i = 1, \dots, n$ and let $\boldsymbol{y} = (y(\boldsymbol{x}_1^*), \dots, y(\boldsymbol{x}_n^*))^T$ denote the set of field observations. We combine the field observations with the digital twin outcomes into the joint output vector $\boldsymbol{z} = (\boldsymbol{y}^T, \boldsymbol{\eta}^T)^T$ [43]. According to the field observation model in Eq. (1) and the GP models in Eqs. (2) and (4), conditioned on the optimal calibration parameter $\boldsymbol{t}^*$ and all other GP parameters, the joint output follows a multivariate Gaussian distribution with a zero mean and a covariance matrix

$$\Sigma_z = \begin{pmatrix} \Sigma_y + \Sigma_\delta & \Sigma_{y,\eta} \\ \Sigma_{y,\eta}^{\mathrm{T}} & \Sigma_\eta \end{pmatrix} + \Sigma_\epsilon, \tag{7}$$

where Σ_y is a $n \times n$ matrix with its (i,j)-th equal to $k_\eta\left((\boldsymbol{x}_i^*, \boldsymbol{t}^*), \left(\boldsymbol{x}_j^*, \boldsymbol{t}^*\right)\right)$, Σ_δ is a $n \times n$ matrix with its (i,j)-th equal to $k_\delta(\boldsymbol{x}_i^*, \boldsymbol{x}_j^*)$, Σ_ϵ is a $(n+m) \times (n+m)$ diagonal matrix with elements ϵ, and $\Sigma_{y,\eta}$ is a $n \times m$ matrix with its (i,j)-th equal to $k_\eta((\boldsymbol{x}_i^*, \boldsymbol{t}^*), (\boldsymbol{x}_j, \boldsymbol{t}_j))$. Consequently, the conditional likelihood is

$$L(\boldsymbol{z} | \alpha_\eta, \boldsymbol{\beta}_\eta, \alpha_\delta, \boldsymbol{\beta}_\delta, \boldsymbol{t}^*, \Sigma_\epsilon) \propto |\Sigma_z|^{-\frac{1}{2}} \exp\left\{ -\frac{1}{2} \boldsymbol{z}^T \Sigma_z^{-1} \boldsymbol{z} \right\}. \tag{8}$$

We can derive the predictive distribution of the calibrated digital twin outcome for an unobserved control parameter set $\boldsymbol{x}_{new}^*$ according to Bayesian statistics as follows [38, 39, 41]:

$$\hat{y}(\boldsymbol{x}_{new}^*) = \Sigma_{\tilde{z}}^T \Sigma_z^{-1} \boldsymbol{z}, \tag{9}$$

$$\hat{\sigma}^2(\boldsymbol{x}_{new}^*) = k((\boldsymbol{x}_{new}^*, \boldsymbol{t}^*), (\boldsymbol{x}_{new}^*, \boldsymbol{t}^*)) - \Sigma_{\tilde{z}}^T \Sigma_z^{-1} \Sigma_{\tilde{z}}, \tag{10}$$

where $\Sigma_{\tilde{z}}$ is a $(n+m)$-vector of covariances between every training input and $\boldsymbol{x}_{new}^*$ with its i-th equal to the sum of $k_\eta((\boldsymbol{x}_i^*, \boldsymbol{t}^*), (\boldsymbol{x}_{new}^*, \boldsymbol{t}^*))$ and $k_\delta(\boldsymbol{x}_i^*, \boldsymbol{x}_{new}^*)$ for $i = 1, \dots, n$ and its j-th equal to $k_\eta((\boldsymbol{x}_j, \boldsymbol{t}_j), (\boldsymbol{x}_{new}^*, \boldsymbol{t}^*))$ for $j = n+1, \dots, n+m$.

3.3. Modular Bayesian calibration

A fully Bayesian calibration simultaneously estimates all parameters. However, this is complicated for high-dimensional parameters and can suffer from unexpected contamination where an unreasonably estimated parameter negatively influences other parameters [44, 45]. To address these challenges, this paper uses a modular approach that functionally divides the Bayesian calibration described in Section 3.2 into five modules. The modularization also enhances algorithmic understanding and flexibility for practitioners [18, 35, 46].

Module 1 estimates the kernel parameters of a digital twin surrogate $(\alpha_\eta, \boldsymbol{\beta}_\eta)$ by incorporating both prior knowledge and digital twin outcomes $\boldsymbol{\eta}$. We use the Metropolis-Hastings (MH) sampling to generate samples from a multivariate distribution, which enables efficient exploration of a high-dimensional parameter space [47]. This enables sequential parameter realizations within a Bayesian inference framework, thereby providing uncertainty quantification [47–49]. A key feature of MH sampling is balancing prior knowledge and data-driven likelihood estimation in Eq. (6). When field and digital twin data are insufficient to estimate the underlying functions of the digital twin model and its discrepancy, the posterior distributions are heavily influenced by their prior distributions. Conversely, with sufficient data, the posterior distributions gradually converge toward the likelihood function, reducing dependency on the prior. Appendix A details the MH-based estimation process for digital twin kernel parameters.

Module 2 calculates Σ_η by the kernel function for the digital twin surrogate in Eq. (3) with the posterior medians of $(\alpha_\eta, \boldsymbol{\beta}_\eta)$. Module 3 estimates the calibration parameters and the kernel parameters of a discrepancy surrogate. MH sampling infers the posterior distributions of the parameters by incorporating both prior knowledge and the likelihood of the joint output values $\boldsymbol{z}$ in Eq. (8). Appendix B details the estimation process. Module 4 calculates Σ_z in Eq. (7) with the observation covariance matrix Σ_ϵ and the posterior medians of $(\alpha_\eta, \boldsymbol{\beta}_\eta, \boldsymbol{t}^*, \alpha_\delta, \boldsymbol{\beta}_\delta)$ by Eqs. (3) and (5). Module 5 performs the calibrated prediction for an unobserved set of control parameters in Eqs. (9) and (10). Fig. 2 shows an overview of the proposed approach highlighting the inputs and outputs for the modules.

4. Calibration framework for digital twin-based decision-making in an AMHS

This section introduces a calibration framework for digital twin-based decision-making in an AMHS (See Fig. 3). Section 4.1 determines a target AMHS and its digital twin model. Section 4.2 specifies a material handling environment of the AMHS. Section 4.3 defines the calibration problem for digital twin-based decision-making in the AMHS. Section 4.4 numerically investigates the prediction error of the model for the digital twin-based decision-making.

4.1. Determination of a field system and its digital twin

Although historical observations from a field system and digital twin outcomes are required for digital twin calibration, field observations are confidential in most companies. Therefore, two simulation models with heterogeneous commercial software are used to represent a field system and its digital twin model. The simulation implemented in Automod from Applied Materials, following guidance from a large electronics manufacturer, is assumed to be an actual field system without additional tuning or operational adjustments, treating it as a black-box representation [50]. The other simulation model implemented in Plant Simulation from Siemens serves as the digital twin model. The digital twin model follows key configurations of the field model as described in Section 4.2. The field model exhibits higher fidelity than the digital twin model, as it incorporates more detailed representations of fabrication production environments, including machine breakdowns and

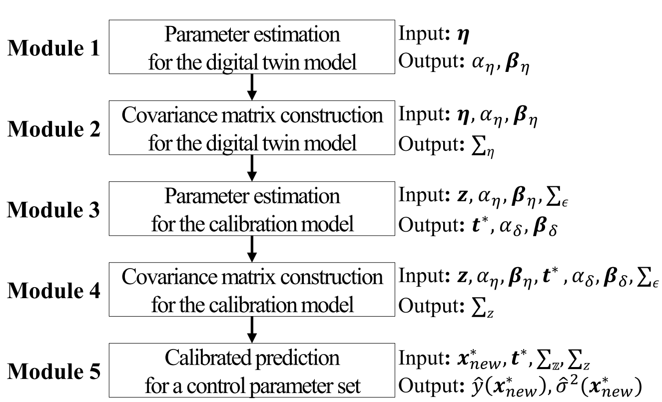

Fig. 2. Overview of the modular Bayesian calibration for digital twin models.

work-in-process control logic. Therefore, these models may exhibit different behaviors due to differences in modeling capabilities and implementation, as well as uncertainties in code and parameters.

4.2. Setup of material handling environment

When the AMHS experiences a significant environmental change, its digital twin model should be recalibrated with updated field and digital twin data. Therefore, a specific material handling environment needs to be defined before the calibration is applied to real-world applications. We consider the AMHS consisting of 20 intra-bays as shown in Fig. 4. An inter-bay has 60 loading/unloading stations for a particular process of the eight processes, e.g., cleaning, etching, photolithography, and diffusion processes. Intra-bay, inter-bay, and outer loops allow an automated guided vehicle (AGV) to deliver a transfer request from the origin to the destination. The origin and destination of a transfer request are determined based on the transaction and stationary matrices driven by the production sequence of a prototype product. Fig. 4 presents the layout configuration of the AMHS.

Three key operational policies are used to control the AGV movements: AGV dispatching, AGV routing, and idle AGV repositioning, as shown in Fig. 5. The dispatching policy selects which AGV dispatches an incoming transfer request. An AGV is dispatched to a transfer request with the closest distance among those in the waiting queue as shown in Fig. 5(a). A routing policy determines the set of nodes to minimize the total travel costs (See Fig. 5(b)). The policy is used to guide an AGV to the route with the shortest travel distance. The repositioning policy determines the dwelling point of an idle AGV for a prompt reaction to the upcoming transfer request. An idle AGV is circulated in the intra-bay, where the AGV completes its transfer request until the AGV dispatches an upcoming transfer request (See Fig. 5(c)).

The operational assumptions in the AMHS were considered as follows: (1) inter-arrival time between transfer requests follows an exponential distribution; (2) production schedules always correspond to loading and unloading operations; (3) loading and unloading operations deterministically require a fixed amount of time; and (4) the maximum speeds of AGVs on straight rail and curved edge are 3 m/s and 1 m/s, respectively.

4.3. Problem definition of digital twin-based decision-making

Digital twin-based decision-making in AMHSs covers the optimization of material handling productivity as well as reducing operating costs, particularly in AGV fleet sizing and specification design problems [1, 51–56]. This paper investigates system performance under various transportation volumes by considering three control parameters: the number of AGVs, the inter-arrival time between transfer requests, and loading/unloading time. The DT, a key performance metric, is the total time to complete a transfer request, which is the sum of the lot awaiting time, loading/unloading time, empty travel time, loaded travel time, and blocking time [57].

AGVs are the resources that fulfill transfer requests. Increasing the number of AGVs decreases their utilization rate, which in turn reduces waiting times and subsequently decreases DT. Conversely, as the inter-arrival time of transfer requests decreases, transportation demand increases, resulting in higher AGV utilization and longer DT. Additionally, a reduction in AGV loading/unloading time directly decreases DT. Since DT changes nonlinearly with these three factors, predicting it is highly challenging.

The parameter uncertainty in a field system exists and the setting of the calibration parameters can substantially impact the digital twin outcomes. For example, the acceleration of a vehicle may vary slightly because of the state of the vehicle or rail or the difference in sensor performance, but the same specification value is typically used as the default setting value in a digital twin model. Based on interviews with industry experts, the acceleration of AGVs and the minimum distance

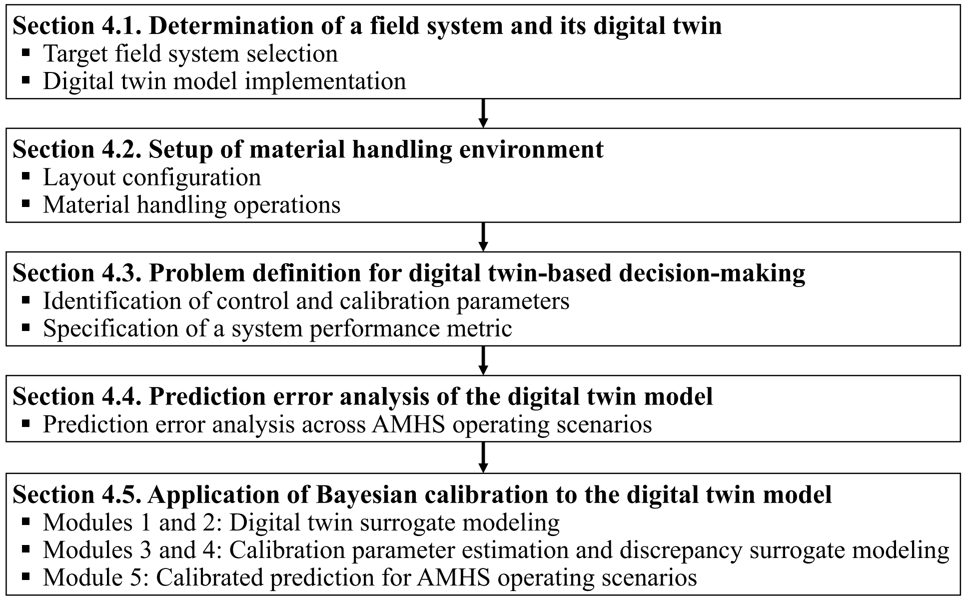

Fig. 3. Calibration framework for digital twin-based decision-making in an AMHS.

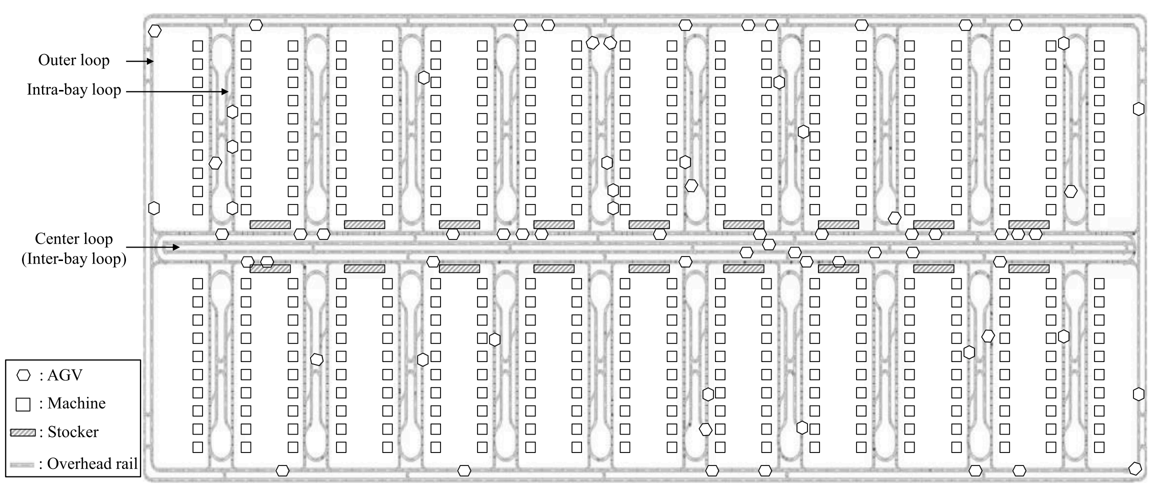

Fig. 4. Layout configuration of the AMHS.

between AGVs, often referred to as vehicle safety distance, were identified as key calibration parameters. Table 1 lists the control and calibration parameters. The digital twin model was used to predict the DT of the field model for 12 hours after confirming that the statistics of both simulations slightly vary from the warm-up period of 12 hours for the digital twin-based decision-making.

4.4. Prediction error analysis of the digital twin model

Since the impact of parameter uncertainty and bias of the digital twin model can vary across a broad range of control parameters, a sensitivity analysis was conducted using the one-factor-at-a-time method for both models with the paired control parameters to assess the prediction error of the uncalibrated digital twin model. The nominal values of the calibration parameters in the digital twin model are set based on those of the field model. Each experiment was repeated ten times to capture the stochastic behavior of the AMHSs. Fig. 6 presents the boxplots for the DTs and the vehicle utilization rates of both models.

Over the range of each control parameter, the variability of the field model exceeds that of the digital twin model by 86.71 %, 90.28 %, and 89.81 %. When the vehicle utilization rate exceeds 85 %, the variability of the field model is higher by 82.52 %, 89.77 %, and 91.67 %. These high operational fluctuations may arise from the difference in model fidelity as the digital twin model fails to capture some intrinsic stochastic operations of the field model. In contrast, when the vehicle utilization rate is below 85 %, both models exhibit stable performance as idle vehicles are available to absorb operational fluctuations.

The mean absolute percentage errors (MAPEs) of the digital twin model for the three control parameters are 3.83 %, 9.03 %, and 5.56 %, respectively. The inter-arrival time between transfer requests significantly affects the MAPE of the digital twin model ranging from 0.37 % to 31.22 %. Meanwhile, the number of AGVs affects the MAPE within a range of 0.03–8.92 %. On average, the digital twin model showed an absolute error of 2.36 % when the field model operates under the vehicle utilization rate below 85 % for all the control parameters. However, when the field model operates under the vehicle utilization rate exceeding 85 %, the digital twin model exhibits a high absolute error of 9.91 %.

The sensitivity analysis reveals the following findings: (1) the field model exhibits greater stochasticity in material handling than the digital twin model; (2) the prediction errors of the digital twin model vary across different control parameters; and (3) the vehicle utilization rate

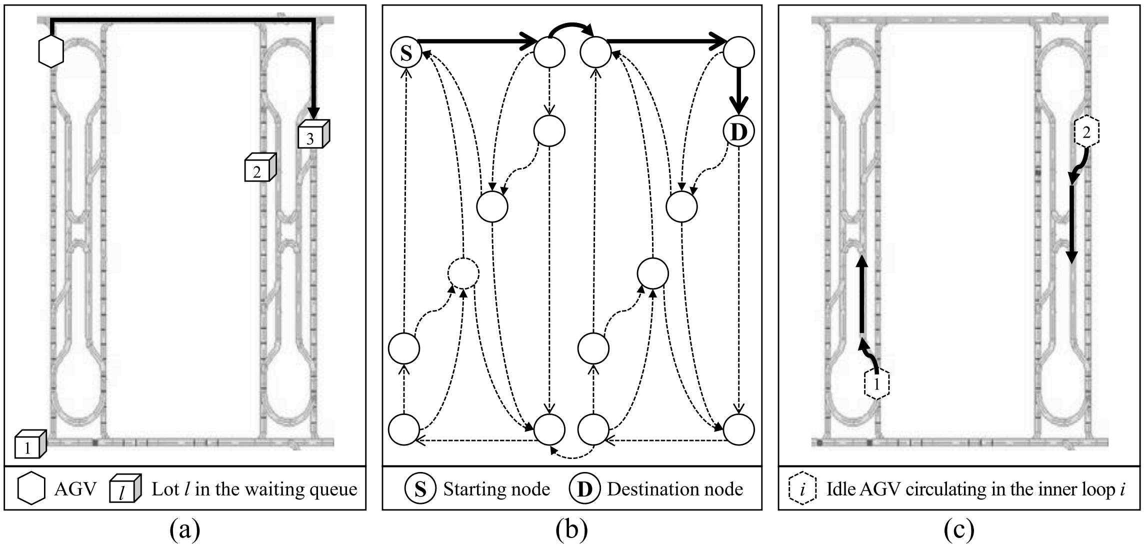

Fig. 5. Illustration of three operational policies for AGVs: (a) AGV dispatching; (b) AGV routing; and (c) idle AGV repositioning.

Table 1
Control and calibration parameters for the digital twin-based decision-making.

	Description	Range		Nominal value
		Min	Max	
Control parameters	Number of AGVs	190	210	200
	Inter-arrival time between transfer requests (s)	0.75	0.90	0.82
	Loading and unloading time (s)	7.50	13.00	10
Calibration parameters	Acceleration of AGVs (m/s^2)	0.50	2.50	1.50
	Minimum distance between AGVs (m)	0.25	2.00	1.12

significantly affects prediction accuracy of the digital twin model. These results highlight the necessity of calibrating the digital twin model for accurate digital twin-based decision-making.

4.5. Application of Bayesian calibration to the digital twin model

The preliminary study shows that the prediction errors in DTs of the digital twin model deviate from material handling conditions (See Fig. 6). To evaluate the Bayesian calibration for various AMHS operating scenarios, we separately considered light-workload scenarios (vehicle utilization rates below 85 %) and heavy-workload scenarios (vehicle utilization rates above 85 %). For training and validation data, the Latin hypercube sampling is used to ensure coverage of the entire response surface with the sampled data [58]. We set the number of field observations to a small number, $n = 5$, emulating the limited availability of field observations in practice. Meanwhile, we set the number of digital twin outcomes to a larger number, $m = 30$, for training the surrogate model for each scenario. One hundred validation data were considered for each scenario. The min-max normalization method was applied to control and calibration parameters. The outcomes were standardized during calibration.

For each scenario, we apply the modular Bayesian calibration to the AMHS digital twin model in a step-by-step manner. The i-th observed DT from the field model is characterized by a vector of the three control parameters $\boldsymbol{x}_i$ and detnoted as $y(\boldsymbol{x}_i)$. The j-th DT outcome from the digital twin model is defined as $y(\boldsymbol{x}_j, \boldsymbol{t}_j)$ where $\boldsymbol{t}_j$ represents a vector of the two calibration parameters. According to Eq. (1), the DT of the field system for an unobserved AMHS operating scenario $\boldsymbol{x}_{new}$ can be calculated

$$y(\boldsymbol{x}_{new}) = \eta(\boldsymbol{x}_{new}, \boldsymbol{t}^*) + \delta(\boldsymbol{x}_{new}) + \epsilon. \tag{11}$$

We use GP priors to draw the predictive response surfaces of both the digital twin model and its discrepancy function as shown in Eqs. (2) and (4). For the digital twin surrogate, Module 1 estimates the digital twin kernel parameters α_η and $\boldsymbol{\beta}_\eta = (\beta_{\eta,1}, \beta_{\eta,2}, \beta_{\eta,3}, \beta_{\eta,4}, \beta_{\eta,5})$ as shown in Eq. (3). α_η represents the overall variance and $\beta_{\eta,1}, \beta_{\eta,2}$, and $\beta_{\eta,3}$ denote the dependence strengths for the three control parameters as the number of AGVs, the inter-arrival time between transfer requests, and the loading/unloading time. $\beta_{\eta,4}$ and $\beta_{\eta,5}$ denote the dependence strengths for the two calibration parameters as the acceleration of AGVs and the minimum distance between AGVs. Then, the 30×30 digital twin covariance matrix Σ_η and the 5×5 field covariance matrix Σ_y are computed in Module 2.

Module 3 simultaneously estimates the calibration parameters $\boldsymbol{t}^* = (t_1^*, t_2^*)$ and the discrepancy kernel parameters α_δ and $\boldsymbol{\beta}_\delta = (\beta_{\delta,1}, \beta_{\delta,2}, \beta_{\delta,3})$ in Eq. (5). t_1^* and t_2^* represent the optimal calibration parameters as the acceleration of AGVs and the minimum distance between AGVs. α_δ represents the overall variance of the discrepancy surrogate. $\beta_{\delta,1}, \beta_{\delta,2}$, and $\beta_{\delta,3}$ denote the dependence strengths of the discrepancy surrogate for the three control parameters as the number of AGVs, the inter-arrival time between transfer requests, and the loading/unloading time, respectively. As illustrated in Appendix A and B, Module (1) and (3) estimate the posterior distributions of the kernel and calibration parameters. Then, Module 4 builds the 5×5 discrepancy covariance matrix Σ_δ and the 5×30 covariance matrix $\Sigma_{y,\eta}$.

Based on the constructed covariance matrices $\Sigma_\eta, \Sigma_y, \Sigma_\delta$, and $\Sigma_{y,\eta}$ from the four modules, Module 5 specifies the 35×35 joint covariance matrix Σ_z according to Eq. (7) and provides calibrated predictions following Eqs. (9) and (10) for evaluating and optimizing the material handling process.

5. Experimental results

To investigate the prediction performance of the calibrated digital twin surrogate $\hat{y}(\boldsymbol{x})$, we compare its accuracy with two benchmarks: (1) the field-only surrogate trained with only field observations $\tilde{y}(\boldsymbol{x})$; (2) the baseline digital twin surrogate using true calibration parameters from the field model without discrepancy compensation $\hat{\eta}(\boldsymbol{x}, \bar{\boldsymbol{t}})$, which emulates minimal parameter uncertainty in the field system. The MH for parameter estimation was set to generate 1000,000 samples including a burn-in period of 100,000 samples. Every 100th chain was taken to

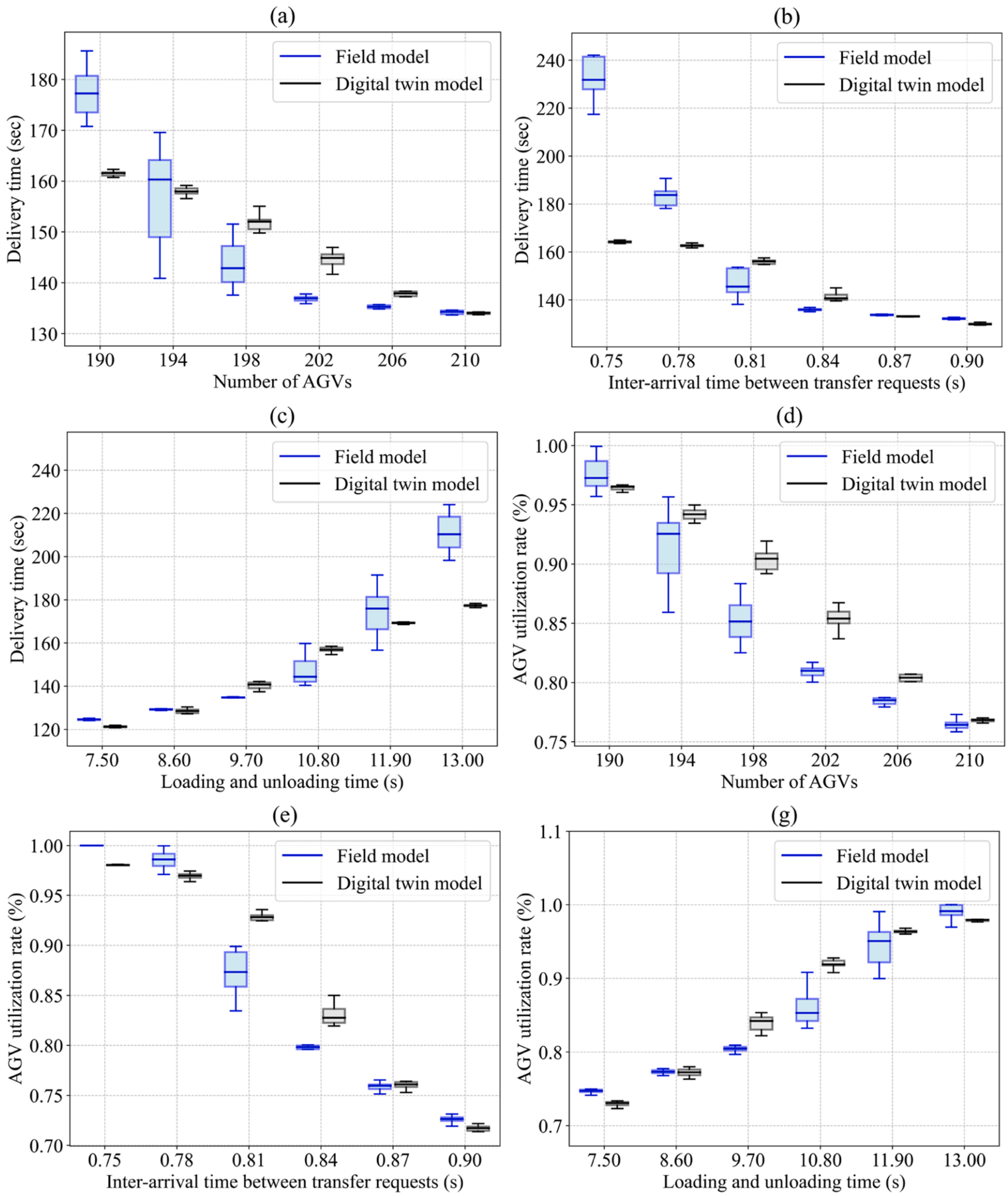

Fig. 6. Boxplots for the AMHS performances from the field model and the digital twin model: (a) DTs with different numbers of AGVs; (b) DTs with different inter-arrival times; (c) DTs with different loading and unloading times; (d) vehicle utilization rates with different numbers of AGVs; (e) vehicle utilization rates with different inter-arrival times; and (f) vehicle utilization rates with different loading and unloading times (color available online).

obtain independent chains. Although some guidelines exist for tuning the hyperparameters, MH samplers often rely on an empirical method through trial and error to achieve efficient parameter selection [59]. Appendix A and B provide detailed procedures for hyperparameter tuning. The experiments were conducted on a PC equipped with the Windows 10 Pro operating system, 32 GB of RAM, and Intel Core i5–14600K @3.50 GHz.

5.1. Digital twin calibration for light-workload scenarios

The MH algorithm was employed to construct a Markov chain for estimating the kernel parameters of the digital twin model, (α_η,$\beta_{\eta,1}$,$\beta_{\eta,2}$, $\beta_{\eta,3}$,$\beta_{\eta,4}$,$\beta_{\eta,5}$). Fig. 7 shows the posterior distributions of the parameters with their prior distributions. The posterior medians were (2.66, 0.04, 0.56, 1.86, 1.73, 1.37). The posterior distributions deviate from the prior distributions, suggesting that the inference is primarily driven by the likelihood rather than the prior. The posterior median of $\beta_{\eta,3}$ suggested that the loading/unloading time is the most influential factor in the digital twin outcomes. The posterior median of $\beta_{\eta,1}$ indicated that the number of AGVs has the most marginal impact on the digital twin outcomes. The estimated $\beta_{\eta,2}$, $\beta_{\eta,4}$, and $\beta_{\eta,5}$ revealed that the inter-arrival time between transfer requests, the acceleration of AGVs, and the minimum distance between AGVs have moderate effects. Managerial

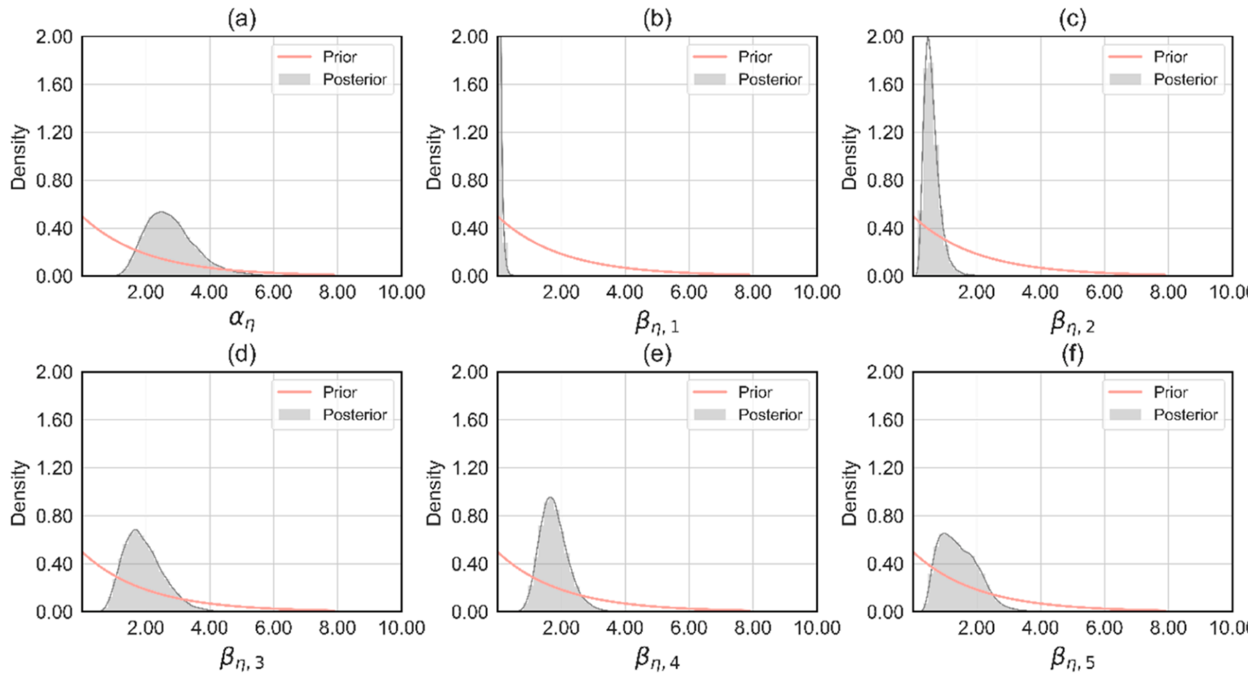

Fig. 7. Prior and posterior distributions of the kernel parameters for the digital twin surrogate with light-workload scenarios: (a) the overall variance; (b) the dependence strength of the number of AGVs; (c) the dependence strength of the inter-arrival time between transfer requests; (d) the dependence strength of the loading and unloading time; (e) the dependence strength of the acceleration of AGVs; and (f) the dependence strength of the minimum distance between AGVs.

findings when the AMHS operates under light-workload scenarios are summarized that: (1) the DT is primarily determined by the duration of vehicles spent on loading/unloading operations; (2) the additional use of AGVs may not enhance the material handling performance.

The MH sampling was employed to estimate the calibration parameters and the discrepancy kernel parameters, $(t_1, t_2, \alpha_\delta, \beta_{\delta,1}, \beta_{\delta,2}, \beta_{\delta,3})$. Fig. 8 shows the posterior distributions of the kernel and calibration parameters alongside their prior distributions. The posterior distributions align more closely with their priors than those of the digital twin kernel parameters, suggesting that the inference relies more on the prior. The posterior medians were (1.54, 1.63, 1.03, 1.36, 1.65, 1.65). The parameter estimation results revealed the following insights: (1) the overall variance of the digital twin surrogate α_η significantly exceeds that of the discrepancy surrogate α_δ; (2) the posterior medians of $\beta_{\eta,1}$ and $\beta_{\delta,1}$ confirmed that the number of AGVs has the least impact on the digital twin surrogate and the discrepancy surrogate among the

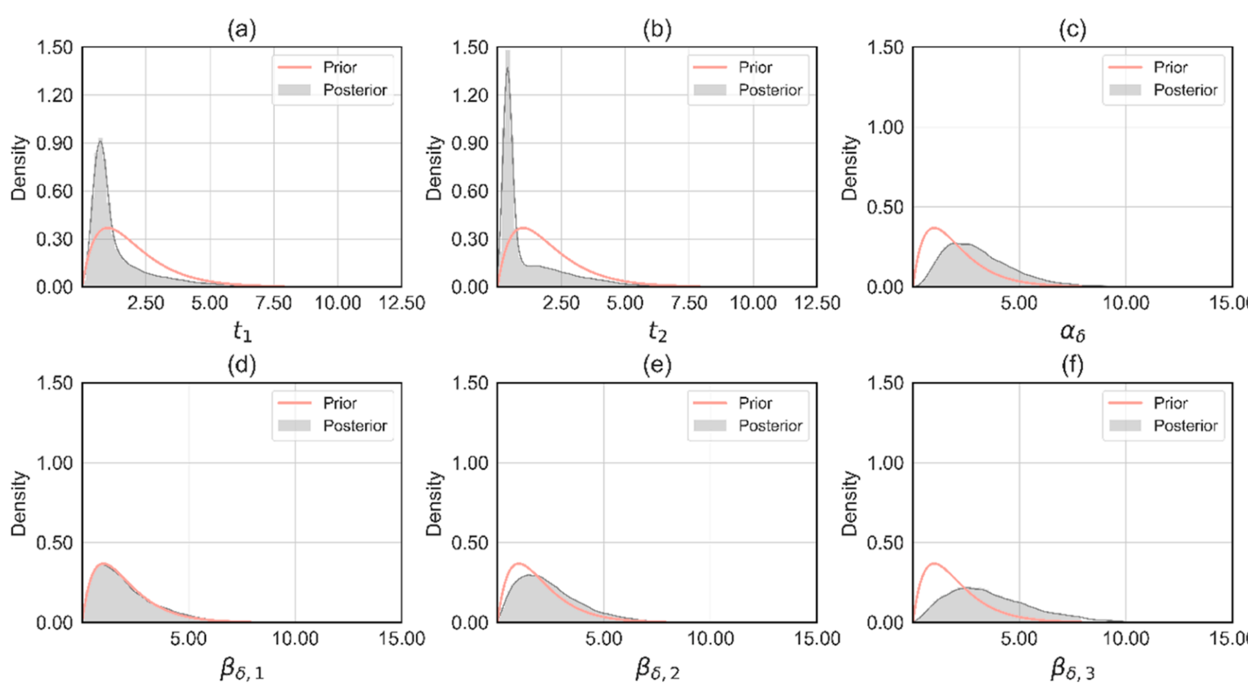

Fig. 8. Prior and posterior distributions of the calibration parameters and the kernel parameters for the discrepancy surrogate with light-workload scenarios: (a) the acceleration of AGVs; (b) the minimum distance between AGVs; (c) the overall variance; (d) the dependence strength of the number of AGVs; (e) the dependence strength of the inter-arrival time between transfer requests; and (f) the dependence strength of the loading and unloading time.

calibration and control parameters.

Three prediction performance metrics were used: mean absolute error (MAE), mean squared error (MSE), and MAPE. Fig. 9 shows the 90 % confidence intervals of the field-only surrogate $\tilde{y}(\boldsymbol{x})$, the baseline digital twin surrogate $\hat{\eta}(\boldsymbol{x}, \bar{\boldsymbol{t}})$, and the calibrated digital twin surrogate $\hat{y}(\boldsymbol{x})$ for the three prediction performance metrics. The width of each confidence interval represents the uncertainty in the parameter estimation process. The results suggest that: (1) the baseline digital twin surrogate trained with the joint data of field observations and simulation outcomes improves the prediction accuracy over the field-only surrogate but the baseline digital twin surrogate shows high predictive variance with wide confidence intervals; and (2) the parameter tuning and discrepancy compensation slightly yields further improvement in prediction accuracy by prevailing the parameter uncertainty and systematic inconsistency of the baseline digital twin surrogate.

Fig. 10 shows scatter plots of the field observations $y(\boldsymbol{x})$ versus the field-only surrogate $\tilde{y}(\boldsymbol{x})$, the baseline digital twin surrogate $\hat{\eta}(\boldsymbol{x}, \bar{\boldsymbol{t}})$, and the calibrated digital twin surrogate $\hat{y}(\boldsymbol{x})$, respectively. The plots convey the prediction tendencies of the surrogate alternatives for validation data. The posterior medians of the calibration parameters and the kernel parameters were used for the predictions. The field-only surrogate with a scant number of training data fails to interpolate a broad range of control parameters. The baseline digital twin surrogate follows the trend of DT with different settings of control parameters but this often underestimates or overestimates the field observations. The proposed approach closely matches their field observations by calibration parameter adjustment and discrepancy compensation.

The calibrated digital twin surrogate $\hat{y}(\boldsymbol{x})$ significantly improves the three performance metrics by 59.58 %, 81.12 %, and 59.13 % compared to the field-only surrogate $\tilde{y}(\boldsymbol{x})$. The proposed approach also outperforms the baseline digital twin surrogate $\hat{\eta}(\boldsymbol{x}, \bar{\boldsymbol{t}})$ by 42.15 %, 63.97 %, and 42.12 %. The results show the calibration parameter adjustment and the discrepancy compensation by the proposed approach can bring substantial improvements when the AMHS operates under light-workload scenarios. The detailed results are shown in Table 2.

5.2. Digital twin calibration for heavy-workload scenarios

For heavy-workload scenarios, the kernel parameters of the digital twin surrogate $(\alpha_\eta, \beta_{\eta,1}, \beta_{\eta,2}, \beta_{\eta,3}, \beta_{\eta,4}, \beta_{\eta,5})$ were estimated from Module 1. Fig. 11 shows the posterior distributions of the parameters alongside their prior distributions. The posterior distributions are substantially different from the prior distributions, suggesting that the likelihood dominates the prior. The posterior medians were (3.16, 0.26, 1.27, 1.91, 1.86, 2.56). The posterior medians of $\beta_{\eta,5}$ and $\beta_{\eta,1}$ indicated that the minimum distance between AGVs has the largest impact on the digital twin surrogate, while the number of AGVs has the smallest impact. The estimated $\beta_{\eta,2}, \beta_{\eta,3}$, and $\beta_{\eta,4}$ suggested that the inter-arrival time between the transfer requests, the loading/unloading time, and the acceleration of AGVs moderately affect the digital twin outcomes. The managerial findings about the AMHS under heavy-workload scenarios are summarized as: (1) decreasing the minimum distance between AGVs may alleviate vehicle congestion considerably; (2) the additional use of AGVs may not significantly improve the DT compared to other parameters.

Parameter estimation of $(t_1, t_2, \alpha_\delta, \beta_{\delta,1}, \beta_{\delta,2}, \beta_{\delta,3})$ were conducted for the heavy-workload scenarios. Fig. 12 shows their prior and posterior distributions. The posterior distributions are more closely aligned with their prior distributions than those of the digital twin kernel parameters, suggesting that the inference is primarily driven by the prior. The posterior medians were (0.91, 0.54, 2.83, 1.70, 2.17, 3.32). The estimated dependence strength parameters of the tree control parameter $\beta_{\delta,1}, \beta_{\delta,2}$, and $\beta_{\delta,3}$ indicate that the loading/unloading time has the largest impact on the discrepancy surrogate while the number of AGVs has the smallest impact. The relatively large ratio of the overall variance for the discrepancy surrogate suggests that practitioners require a more sophisticated discrepancy compensation to construct a credible digital twin model when the system experiences heavy workloads.

Fig. 13 shows the 90 % confidence intervals of the field-only surrogate $\tilde{y}(\boldsymbol{x})$, the baseline digital twin surrogate $\hat{\eta}(\boldsymbol{x}, \bar{\boldsymbol{t}})$, and the calibrated digital twin surrogate $\hat{y}(\boldsymbol{x})$ during parameter estimation. Combining field observations and digital twin outcomes leads to prediction performance improvements for estimating the response surface of the field model across a broad range of control parameters. Furthermore, the parameter tuning and discrepancy compensation from Module 3 significantly outperform the baseline digital twin surrogate constructed by Module 1 in the performance metrics. The calibrated digital twin surrogate shows less variance with narrower confidence intervals in the performance metrics than other surrogate alternatives.

Fig. 14 presents the scatter plots of the field observations $y(\boldsymbol{x})$ versus the field-only surrogate $\tilde{y}(\boldsymbol{x})$, the baseline digital twin surrogate $\hat{\eta}(\boldsymbol{x}, \bar{\boldsymbol{t}})$, and the calibrated digital twin surrogate $\hat{y}(\boldsymbol{x})$, respectively. The posterior medians were used for the predictions. The inaccuracies of $\tilde{y}(\boldsymbol{x})$, $\hat{\eta}(\boldsymbol{x}, \bar{\boldsymbol{t}})$, and $\hat{y}(\boldsymbol{x})$ were higher than the light-workload scenarios because the AMHS shows high system variability and dynamics with the change

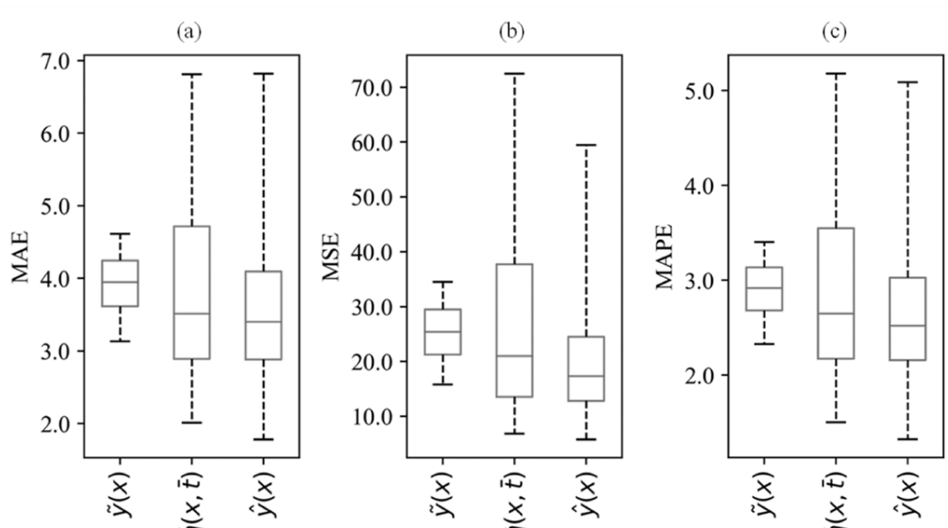

Fig. 9. 90 % confidence levels with 5.00 % and 95.00 % of the surrogate alternatives with light-workload scenarios for the prediction metrics: (a) MAE; (b) MSE; and (C) MAPE.

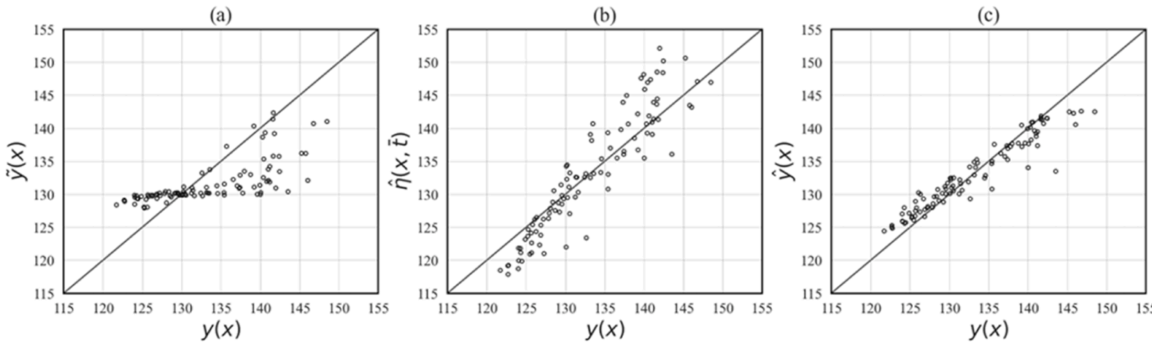

Fig. 10. Comparisons of the DTs (s) between the field model and the surrogate alternatives under light-workload scenarios: (a) plot of $y(\boldsymbol{x})$ versus $\widetilde{y}(\boldsymbol{x})$; (b) plot of $y(\boldsymbol{x})$ versus $\widehat{\eta}(\boldsymbol{x},\overline{\boldsymbol{t}})$; and (c) plot of $y(\boldsymbol{x})$ versus $\widehat{y}(\boldsymbol{x})$.

Table 2
Summary of the prediction performance metrics with light-workload scenarios.

	MAE(s)	MSE(s^2)	MAPE(%)
$\widetilde{y}(\boldsymbol{x})$	4.1800	27.5986	3.0907
$\widehat{\eta}(\boldsymbol{x},\overline{\boldsymbol{t}})$	2.9203	14.4604	2.1820
$\widehat{y}(\boldsymbol{x})$	1.6895	5.2100	1.2630

in control parameters. The operational fluctuations led by the changes heavily affect the congested AMHS performances.

Table 3 lists the numerical results for the three prediction performance measures. The calibrated digital twin surrogate $\widehat{y}(\boldsymbol{x})$ remarkably outperforms the field-only surrogate $\widetilde{y}(\boldsymbol{x})$ in the three performance metrics by 20.75 %, 39.84 %, and 20.68 %. The proposed approach also improves prediction accuracy compared to the baseline digital twin surrogate $\widehat{\eta}(\boldsymbol{x},\overline{\boldsymbol{t}})$ by 4.19 %, 12.70 %, and 4.69 %. The results indicate that the parameter tuning and discrepancy compensation from Module 3 can be crucial to gain further improvements over the baseline digital twin surrogate.

5.3. Applicability to digital twin models across diverse domains

With its strong theoretical foundation, the approach provides a rigorous framework for constructing credible digital twins. This section examines the applicability of the Bayesian calibration to digital twin models across industrial domains. The specific characteristics of digital twin models where the approach is highly applicable include the following:

- **Bias-prone digital twin model in complex environments**: complex systems inherently exhibit digital twin model bias which cannot be entirely addressed through calibration parameter optimization due to the challenges posed by parameter uncertainty. These systems also require extensive programming with code uncertainty to coordinate interactions between machines and workpieces, necessitating Bayesian calibration for discrepancy compensation.
- **Off-line calibration for process evaluation and optimization in dynamic environments**: To support the entire product lifecycle, a digital twin must integrate both performance- and event-level digital twins. The proposed approach enables steady-state process diagnosis through performance-level calibration, by incorporating real-time

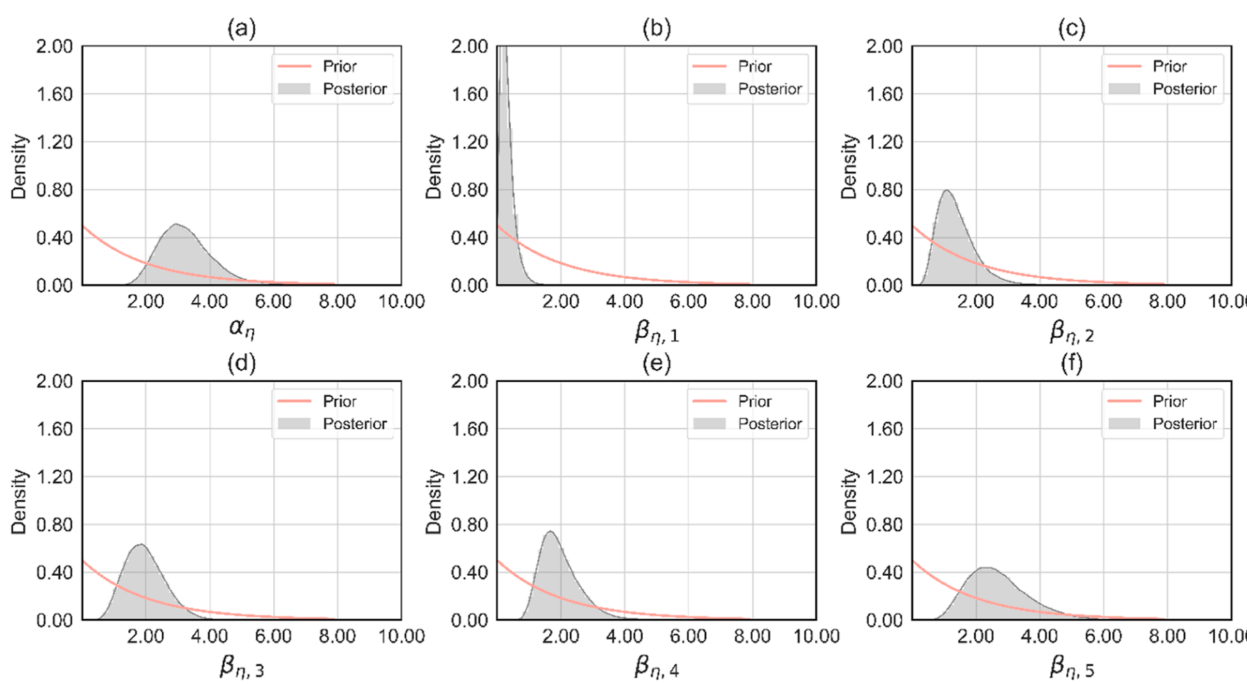

Fig. 11. Prior and posterior distributions of the kernel parameters for the digital twin surrogate with heavy-workload scenarios: (a) the overall variance; (b) the dependence strength of the number of AGVs; (c) the dependence strength of the inter-arrival time between transfer requests; (d) the dependence strength of the loading and unloading time; (e) the dependence strength of the acceleration of AGVs; and (f) the dependence strength of the minimum distance between AGVs.

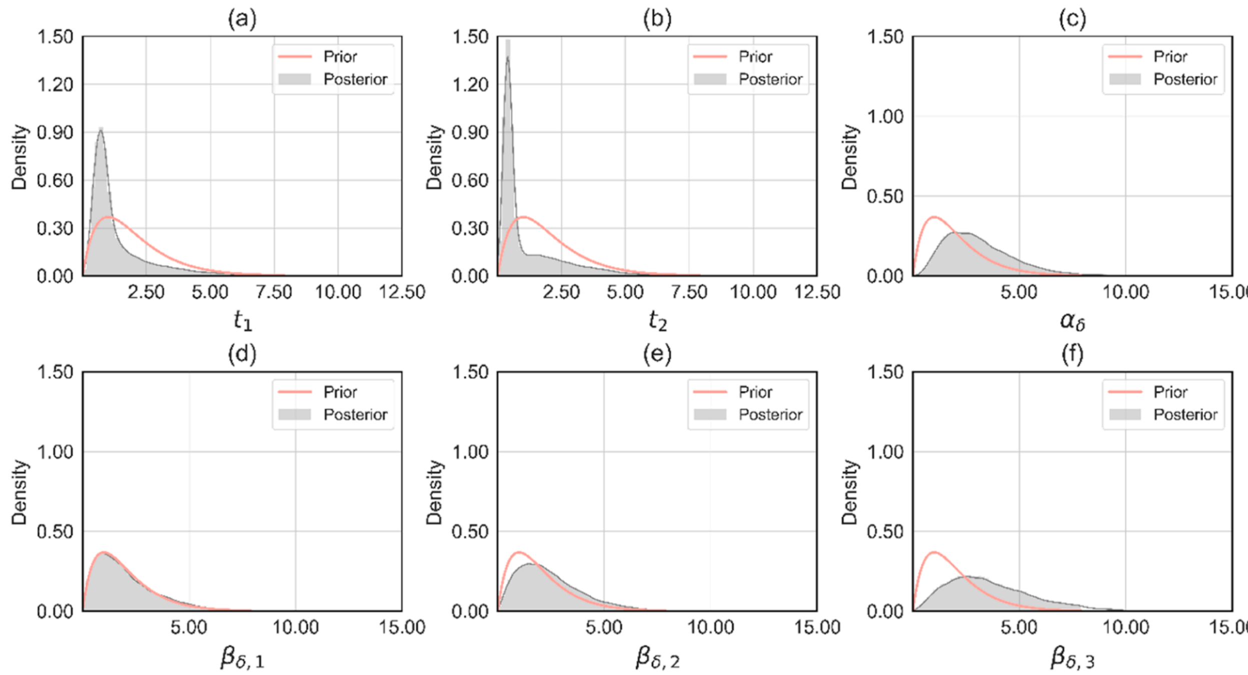

Fig. 12. Prior and posterior distributions of the calibration parameters and the kernel parameters for the discrepancy surrogate with heavy-workload scenarios: (a) the acceleration of AGVs; (b) the minimum distance between AGVs; (c) the overall variance; (d) the dependence strength of the number of AGVs; (e) the dependence strength of the inter-arrival time between transfer requests; and (f) the dependence strength of the loading and unloading time.

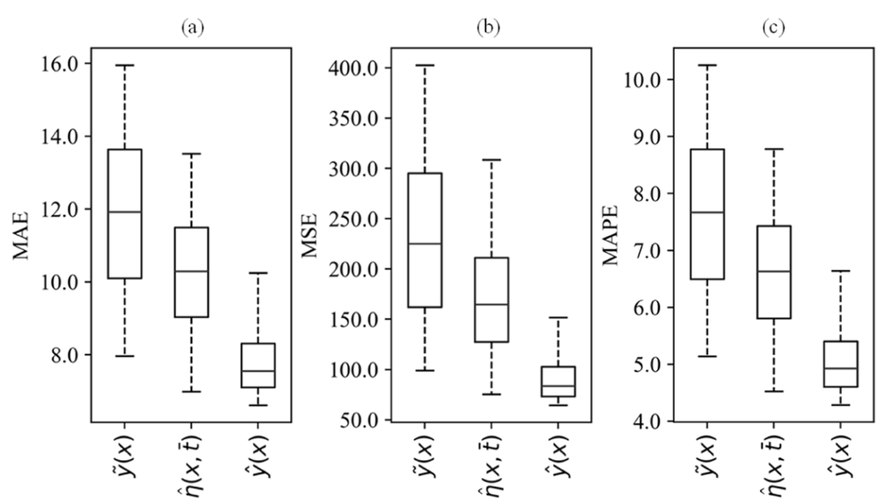

Fig. 13. 90 % confidence levels with 5.00 % and 95.00 % of the surrogate alternatives with heavy-workload scenarios for the prediction metrics: (a) MAE; (b) MSE; and (C) MAPE.

system dynamics, such as time-varying parameters. However, if the system undergoes fundamental behavioral changes, recalibration becomes essential to address non-stationarity and ensure the sustainability of digital twins.

- **Large-scale digital twin model in data-sparse environments**: In general, field data acquisition from large-scale systems is highly constrained, and evaluating high-fidelity digital twin models is also computationally expensive. The presented GP surrogates provide a non-linear and flexible regression and replace a digital twin model and its discrepancy with cheap-to-evaluate stochastic approximations. This enables prompt calibration for stochastic digital twin models using a limited number of data in data-sparse environments.
- **System allowing informative prior knowledge for surrogate models:** MH sampling utilizes prior knowledge to build both digital twin and discrepancy surrogates. Empirical insights regarding the impacts of input and calibration parameters on the surrogates can accelerate digital twin calibration processes. Field systems leveraging IoT, networks, and databases enable the use of operating data to extract prior knowledge, supporting rapid Bayesian calibration in fast-evolving environments.

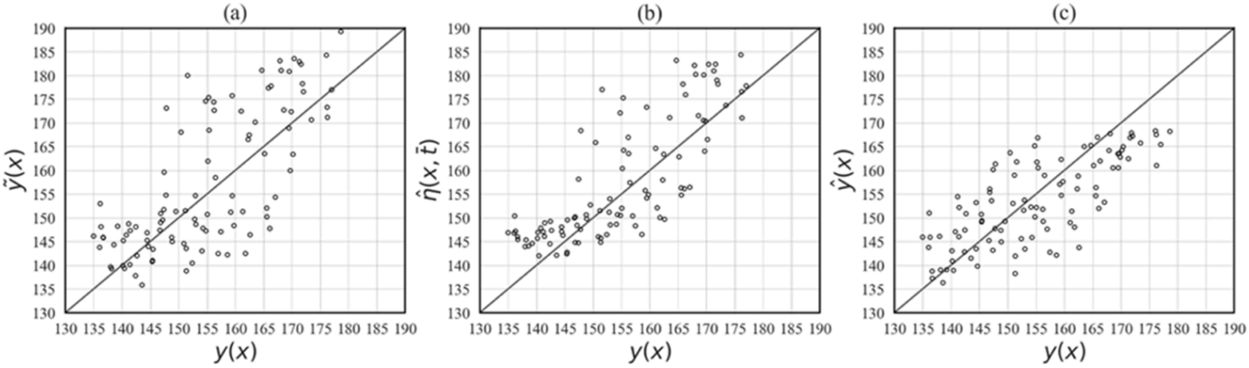

Fig. 14. Comparisons of the DTs (s) between the field model and the surrogate alternatives under heavy-workload scenarios: (a) plot of $y(\boldsymbol{x})$ versus $\widetilde{y}(\boldsymbol{x})$; (b) plot of $y(\boldsymbol{x})$ versus $\widehat{\eta}(\boldsymbol{x}, \bar{\boldsymbol{t}})$; and (c) plot of $y(\boldsymbol{x})$ versus $\widehat{y}(\boldsymbol{x})$.

Table 3
Summary of the prediction performance metrics with heavy-workload scenarios.

	MAE(s)	MSE(s^2)	MAPE(%)
$\widetilde{y}(\boldsymbol{x})$	8.2563	104.0999	5.3057
$\widehat{\eta}(\boldsymbol{x}, \bar{\boldsymbol{t}})$	6.8293	71.7341	4.4157
$\widehat{y}(\boldsymbol{x})$	6.5434	62.6252	4.2085

6. Conclusion

Digital twin models inevitably include parameter uncertainty and discrepancies. For accurate digital twin-based decision-making in AMHSs, this paper proposes a Bayesian calibration. The approach serves as an offline calibration tool enabling a credible digital twin model for process evaluation and optimization to respond to upcoming transportation demand. We also present a digital twin calibration framework to provide practical guidelines and enhance the understanding of AMHS digital twin calibrations for practitioners. The simulation results demonstrate that: (1) the approach properly predicts field system responses across a broad range of control and calibration parameters with a small number of field observations. This also supports rapid calibration for upcoming production schedules in real-world systems since no additional simulation run is required; (2) careful digital twin-based decision-making is necessary when an AMHS experiences heavy workload because of increased uncertainty and irregularity in the material handling processes; and (3) the calibration also offers advantages for strategic AMHS management, providing greater managerial tractability to analyze the impact of control and calibration parameters on digital twin model outcomes and discrepancies.

As calibrating digital twin models has potential for further improvement, we suggest future directions as follows: (1) the presented GP surrogates are concerned with continuous spaces, whereas field systems often involve the constrained and combinatorial nature of problems. Investigating a surrogate model to incorporate operational constraints can improve the adaptability of the approach to extended applications; (2) when a real-world system shows fundamental changes in behavior, a dynamic calibration can be employed to detect non-stationarity and trigger recalibration. The investigation may ensure consistent digital twins over time; (3) the GP surrogates assume the relationship between input parameters and outputs remains smooth and consistent across spatial coordinates. However, digital twin models and their discrepancies may exhibit localized variations in different parameter regions, leading to performance degradation in prediction accuracy. To address this, a kernel function that accounts for spatial discontinuities can be explored; and (4) when prior modeling remains challenging, a systematic prior knowledge elicitation procedure can facilitate faster and more reliable surrogate modeling for the Bayesian calibration.

CRediT authorship contribution statement

Hong Soondo: Writing – review & editing, Supervision, Resources, Investigation, Funding acquisition, Conceptualization. **Kim Haejoong:** Writing – review & editing, Supervision, Investigation, Data curation, Conceptualization. **Park Chiwoo:** Writing – review & editing, Validation, Methodology, Investigation, Conceptualization. **Kang Bonggwon:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation.

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Appendix

A. MH Sampling for digital twin kernel parameters

The MH sampling iteratively realizes the samples of the kernel parameter set for the digital twin surrogate. This approximates the posterior distributions of the kernel parameters by evaluating the likelihood with prior knowledge. Table A. 1 describes the pseudocode of the MH sampling. Through preliminary sensitivity analyses, we consider the gamma and normal distributions for their prior and proposal distributions, respectively.

Table A.1
Pseudocode of the MH sampling for digital twin kernel parameters

Metropolis-hasting sampling for digital twin kernel parameters		
1:	Obtain random point α_η^0 and $\boldsymbol{\beta}_\eta^0$	
2:	**For** each s **do**	
3:		Generate $\overline{\alpha}_\eta$ and $\overline{\boldsymbol{\beta}}_\eta$ from each proposal distribution given α_η^s and $\boldsymbol{\beta}_\eta^s$, respectively
4:		Compute $c = \min\left\{1, \frac{L(\mathbf{z}\mid\overline{\alpha}_\eta, \overline{\boldsymbol{\beta}}_\eta)\pi(\overline{\alpha}_\eta)\pi(\overline{\boldsymbol{\beta}}_\eta)}{L(\mathbf{z}\mid\alpha_\eta^s, \boldsymbol{\beta}_\eta^s)\pi(\alpha_\eta^s)\pi(\boldsymbol{\beta}_\eta^s)}\right\}$
5:		Sample $r \sim U(0,1)$
6:		Set α_η^{s+1} and $\boldsymbol{\beta}_\eta^{s+1}$ as $\overline{\alpha}_\eta$ and $\overline{\boldsymbol{\beta}}_\eta$ if $c \geq r$; otherwise set them as α_η^s and $\boldsymbol{\beta}_\eta^s$
7:	**End**	

The MH sampler requires tuning of prior distributions and the number of chains. Parameter tuning for specific estimation problems often relies on a trial-and-error approach [34]. We use independent gamma priors for the digital twin kernel parameters, $(\alpha_\eta, \beta_{\eta,1}, \beta_{\eta,2}, \beta_{\eta,3}, \beta_{\eta,4}, \beta_{\eta,5})$ [16]. To simplify the hyperparameter tuning process, we use the same value across the digital twin kernel parameters for shape and rate parameters, respectively. The gamma priors, a and b, are defined as follows:

$$\pi(\alpha_\eta) \propto \alpha_\eta^{a-1} e^{-b\alpha_\eta}, \tag{A.1}$$

$$\pi(\beta_{\eta,k}) \propto \beta_{\eta,k}^{a-1} e^{-b\beta_{\eta,k}}, \quad k = 1, \ldots, p+l. \tag{A.2}$$

We conducted a sensitivity analysis by varying the shape and rate parameters from 0.50 to 2.00 in increments of 0.50 to improve the accuracy of calibrated predictions. Based on this analysis, we identified 1.00 and 2.00 as the most effective parameter values across both workload scenarios. We also varied the number of chains from 100,000 to 1000,000 and selected 1000,000 based on visual assessment of trace plots, ensuring adequate mixing and convergence. Fig. A. 1 and Fig. A. 2 present the trace plots. The trace plots of six chains indicate adequate mixing, suggesting effective exploration and convergence of the parameter space.

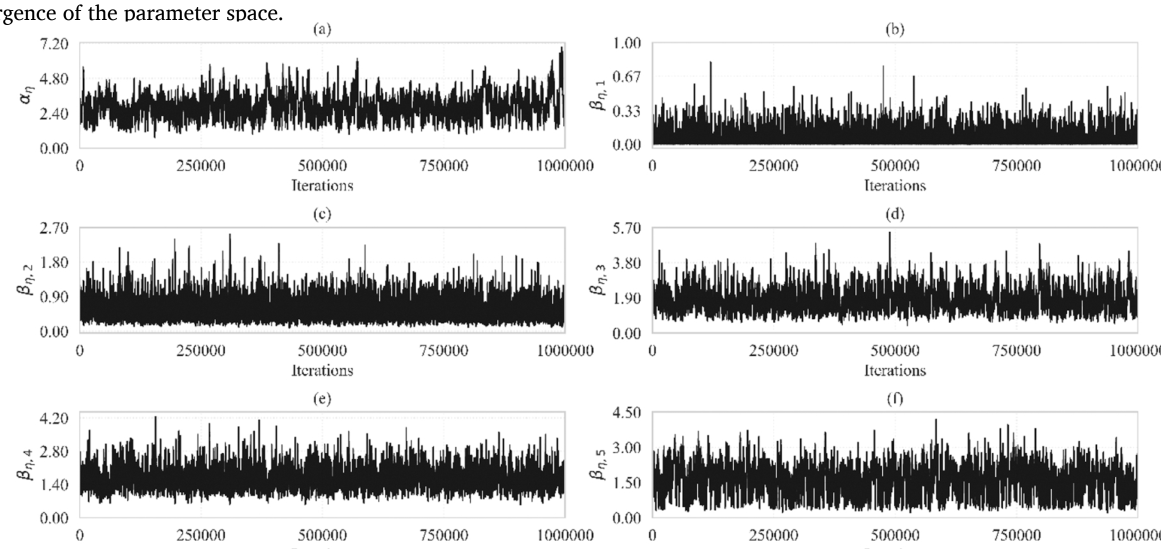

Fig. A.1. Trace plots for the kernel parameters of the digital twin surrogate with light-workload scenarios: (a) the overall variance; (b) the dependence strength of the number of AGVs; (c) the dependence strength of the inter-arrival time between transfer requests; (d) the dependence strength of the loading and unloading time; (e) the dependence strength of the acceleration of AGVs; and (f) the dependence strength of the minimum distance between AGVs

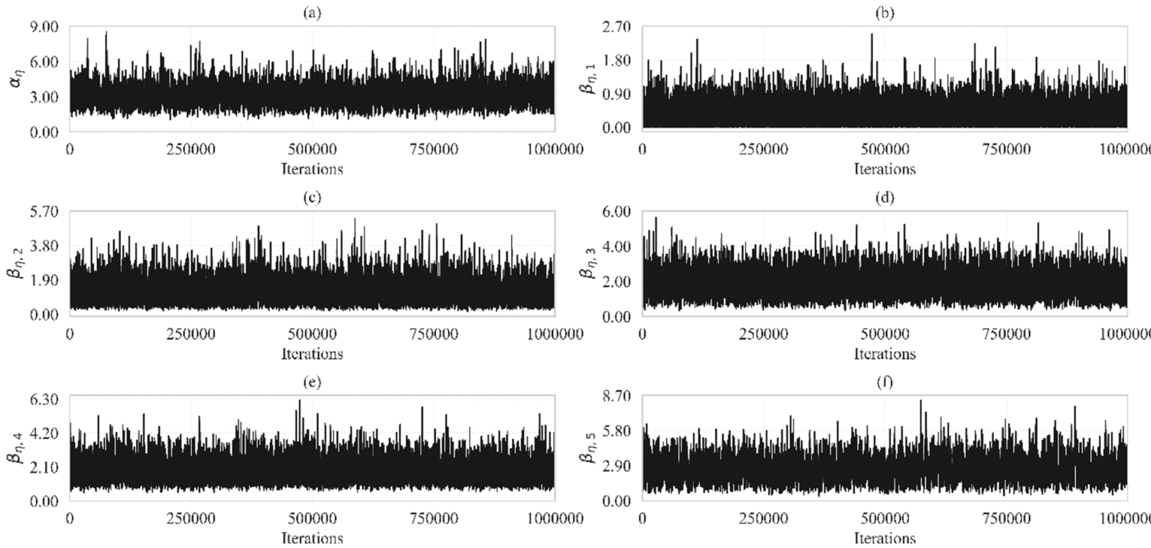

Fig. A.2. Trace plots for the kernel parameters of the digital twin surrogate with heavy-workload scenarios: (a) the overall variance; (b) the dependence strength of the number of AGVs; (c) the dependence strength of the inter-arrival time between transfer requests; (d) the dependence strength of the loading and unloading time; (e) the dependence strength of the acceleration of AGVs; and (f) the dependence strength of the minimum distance between AGVs

B. MH Sampling for calibration parameters and discrepancy kernel parameters

A MH sampling was used to estimate the calibration parameters and the kernel parameters of the discrepancy surrogate. In this stage, the MH sampling iteratively evaluates the probability of the joint output values $\boldsymbol{z}$ and updates the parameter chains. Table A. 2 describes the pseudocode of the MH sampling. After we performed sensitivity analyses based on a trial-and-error manner, the gamma and normal distributions were considered for their prior and proposal distributions, respectively.

Table A.2
Pseudocode of the MH sampling for calibration and discrepancy kernel parameters

Metropolis-hasting sampling for calibration and discrepancy kernel parameters		
1:	Obtain random point $\boldsymbol{t}^0, \alpha_\delta^0$, and $\boldsymbol{\beta}_\delta^0$	
2:	**For** each s **do**	
3:		Generate $\bar{\boldsymbol{t}}, \bar{\alpha}_\delta$, and $\bar{\boldsymbol{\beta}}_\delta$ from each proposal distribution given $\boldsymbol{t}^s, \alpha_\delta^s$, and $\boldsymbol{\beta}_\delta^s$, respectively
4:		Compute $c = \min\left\{1, \frac{L(\boldsymbol{z}\mid\bar{\boldsymbol{t}}, \bar{\alpha}_\delta, \bar{\boldsymbol{\beta}}_\delta)\pi(\bar{\boldsymbol{t}})\pi(\bar{\alpha}_\delta)\pi(\bar{\boldsymbol{\beta}}_\delta)}{L(\boldsymbol{z}\mid\boldsymbol{t}^s, \alpha_\delta^s, \boldsymbol{\beta}_\delta^s)\pi(\boldsymbol{t}^s)\pi(\alpha_\delta^s)\pi(\boldsymbol{\beta}_\delta^s)}\right\}$
5:		Sample $r \sim U(0,1)$
6:		Set $\boldsymbol{t}^{s+1}, \alpha_\delta^{s+1}$, and $\boldsymbol{\beta}_\delta^{s+1}$ as $\bar{\boldsymbol{t}}, \bar{\alpha}_\delta$, and $\bar{\boldsymbol{\beta}}_\delta$ if $c \geq r$; otherwise set them as $\boldsymbol{t}^s, \alpha_\delta^s$, and $\boldsymbol{\beta}_\delta^s$
7:	**End**	

We follow the same prior modeling procedure described in Appendix A for the calibration and discrepancy kernel parameters, $(t_1, t_2, \alpha_\delta, \beta_{\delta,1}, \beta_{\delta,2}, \beta_{\delta,3})$. We assign the same shape and rate parameter values across the calibration and discrepancy kernel parameters. The gamma priors are defined as follows:

$$\pi(t_k) \propto t_k^{a-1} e^{-bt_k}, \quad k = 1, \ldots, l, \tag{A.3}$$

$$\pi(\alpha_\delta) \propto \alpha_\delta^{a-1} e^{-b\alpha_\delta}, \tag{A.4}$$

$$\pi(\beta_{\delta,k}) \propto \beta_{\delta,k}^{a-1} e^{-b\beta_{\delta,k}}, \quad k = 1, \ldots, p+l. \tag{A.5}$$

For both workload scenarios, we set a and b to 2.00 and 1.00, respectively. We also varied the number of chains from 100,000 to 1000,000 and chose 1000,000. Fig. A. 3 and Fig. A. 4 illustrate trace plots of twelve chains, suggesting adequate mixing.

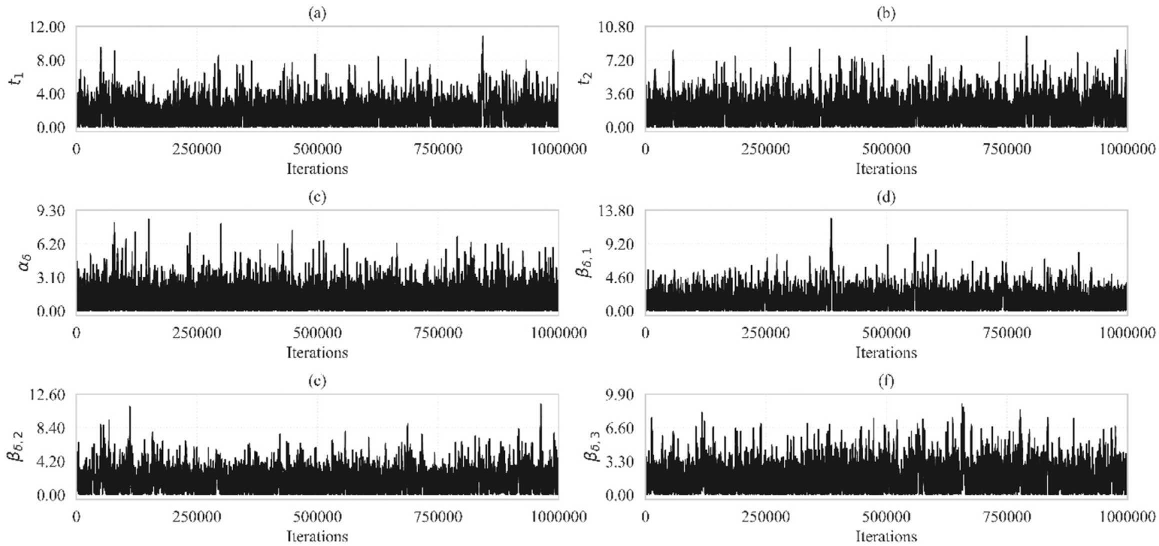

Fig. A.3. Trace plots for the calibration parameters and the kernel parameters of the discrepancy surrogate with light-workload scenarios: (a) the acceleration of AGVs; (b) the minimum distance between AGVs; (c) the overall variance; (d) the dependence strength of the number of AGVs; (e) the dependence strength of the inter-arrival time between transfer requests; and (f) the dependence strength of the loading and unloading time

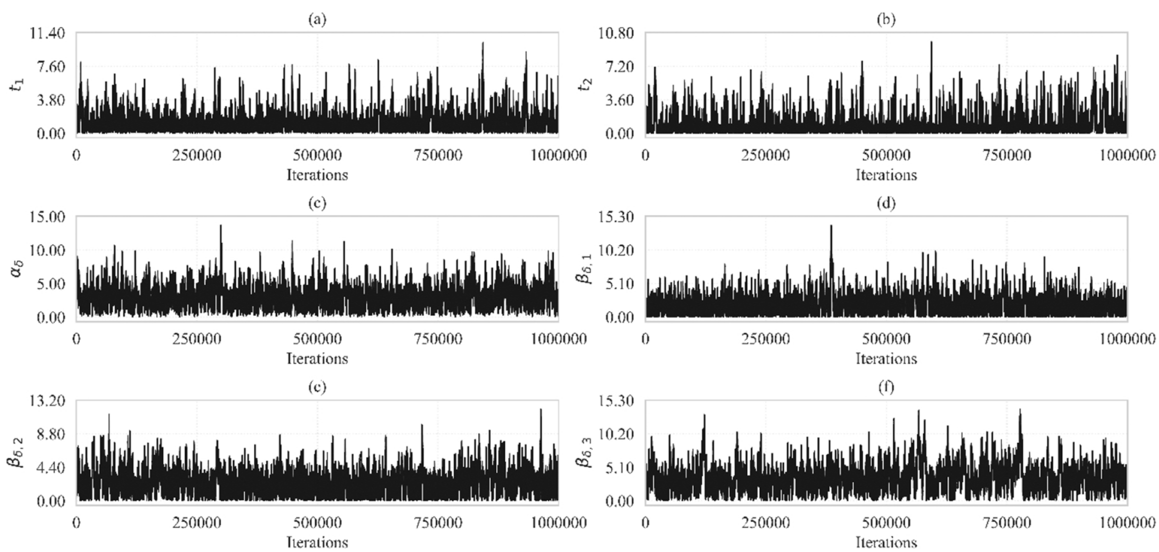

Fig. A.4. Trace plots for the calibration parameters and the kernel parameters of the discrepancy surrogate with heavy-workload scenarios: (a) the acceleration of AGVs (b) the minimum distance between AGVs; (c) the overall variance; (d) the dependence strength of the number of AGVs; (e) the dependence strength of the inter-arrival time between transfer requests; and (f) the dependence strength of the loading and unloading time

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